PROJECT :

Network Design for Faculty of Computing Block N28c

TASK 4: MAKING THE CONNECTION - LAN AND WAN

SECR1213 - NETWORK COMMUNICATIONS

SEMESTER I, SESSION 2024/2025

SECTION 02

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4.1 Identify the work areas on floor plan

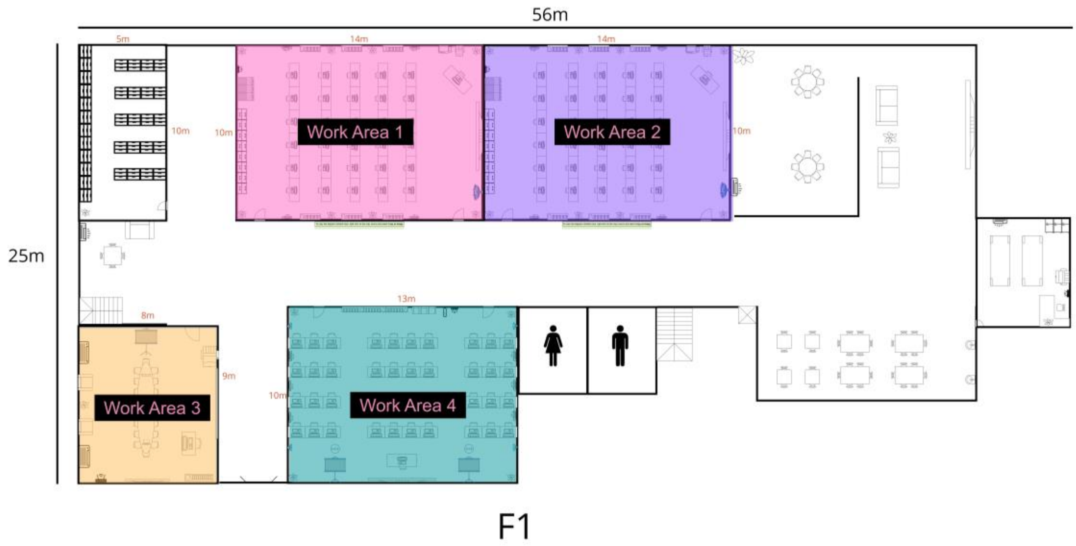

Figure 4.1.1: Work Area at Ground Floor

As shown in Figure 4.1.1, Workspace 1 and Workspace 2 are general-purpose lab designed for use by students in the classroom, each of which is 14 meters long and 10 meters wide. These LABS are equipped with 31 PC workstations and require two 48-port network switches to ensure smooth connectivity of all devices.

The third workspace, located in the lower left corner of the first floor, is a video conference room dedicated to university faculty. This space is equipped with high-quality audiovisual technology to support virtual meetings, lectures and collaboration. It enables faculty to conduct blended and distance courses, interact with colleagues and students from different places, and host guest presentations without the need to travel. The room typically features tools for screen sharing and interactive discussion, promoting flexibility and efficiency for academic and administrative purposes.

Finally, Workspace 4, shown in Figure 4.1.1, is a hybrid classroom with 40 workstations. This environment allows for face-to-face and distance learning, enabling students to participate in real-time classes no matter where they are. Hybrid classrooms integrate technology such as cameras, microphones, and screen-sharing tools to provide students with a seamless experience, physical or virtual. This setting ensures inclusive learning, supports the needs of diverse students, and maintains academic continuity in a variety of contexts.

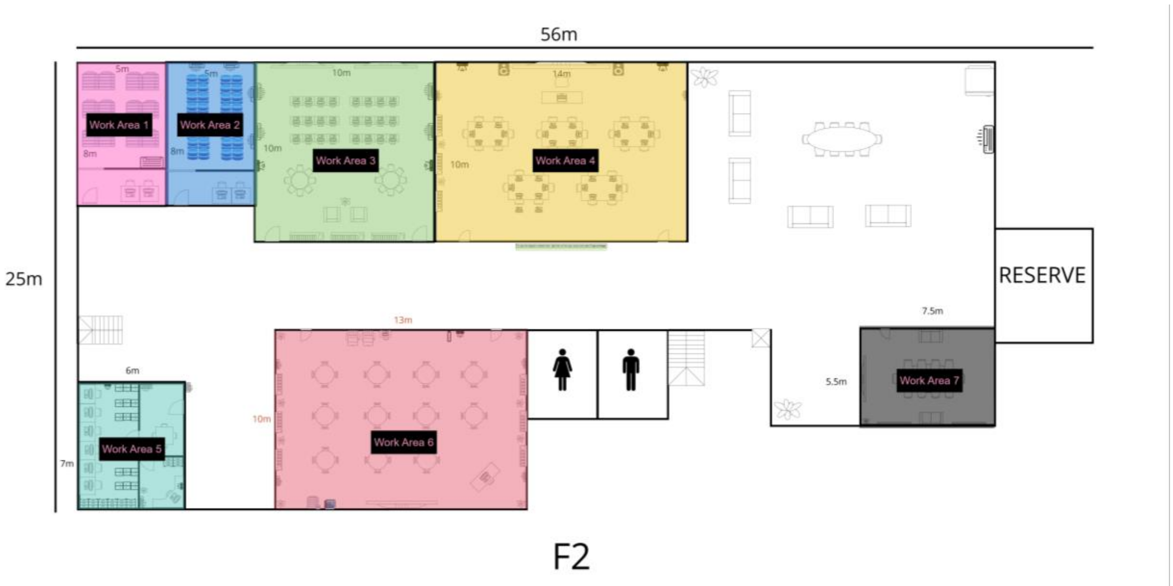

Figure 4.1.2: Work Area at First Floor

Work area 1, the server room, is dedicated to the storage and protection of critical data while implementing the necessary monitoring and security measures. It contains two PC workstations and three servers, all of which are connected using cables to ensure reliable communication and data transfer.

Move to Workspace 2, the Database Room, a space dedicated to managing and analyzing large data sets, often used in academic and research environments. It is equipped with servers and storage systems to securely store university data including research, student records and administrative information. The room supports data processing, collaboration on data-driven projects, and ensures regular backup and recovery to maintain data integrity and accessibility. It also features a PC workstation for related tasks.

In the upper part of the plan (Figure 4.1.2), Workspace 3 is the Network Operations Center (NOC), a centralized hub where IT professionals monitor and manage the university's network infrastructure. NOC ensures the continuous operation of network systems, detects and resolves problems, and maintains optimum performance. It plays a vital role in supporting the campus network, distance learning and communication between departments. The NOC contains 24 workstations, connected via a single 48-port switch, and includes a wireless access point for increased connectivity.

Work Area 4 is the Embedded Lab, a facility dedicated to developing, testing, and prototyping embedded systems, which are custom computing devices designed for specific tasks. The lab provides students with hands-on experience in electronics and robotics, supporting activities such as microcontroller-based application design and prototyping. The embedded lab is equipped with 31 PC workstations, an access point, and a large TV screen. These devices are all connected via 48-port switches to ensure seamless operation.

Work Area 5 is a maintenance room dedicated to the maintenance and repair of IT infrastructure, including network equipment, servers, and other technical systems. It is used to rectify hardware faults, replace faulty components, update software, and maintain the reliability of key systems, such as data centers and communication networks. The room contains seven workstations and a wireless access point connected by cable for stable network access.

As shown in Figure 4.1.1, Workspace 6 on the ground floor is the Cisco Networking Lab, where faculty and students can engage in hands-on learning and skill development related to networking technologies. The lab is equipped with a workstation, a wireless access point and two printers to support practical exercises and activities.

Finally, Workspace 7 is the meeting room, designed as a multi-functional space for productive gatherings and discussions. It is equipped with a large TV, providing a perfect environment for presentations and team collaboration. This versatile room blends comfort and productivity to meet a variety of meeting needs.

4.2 The Distribution and Connection of Devices

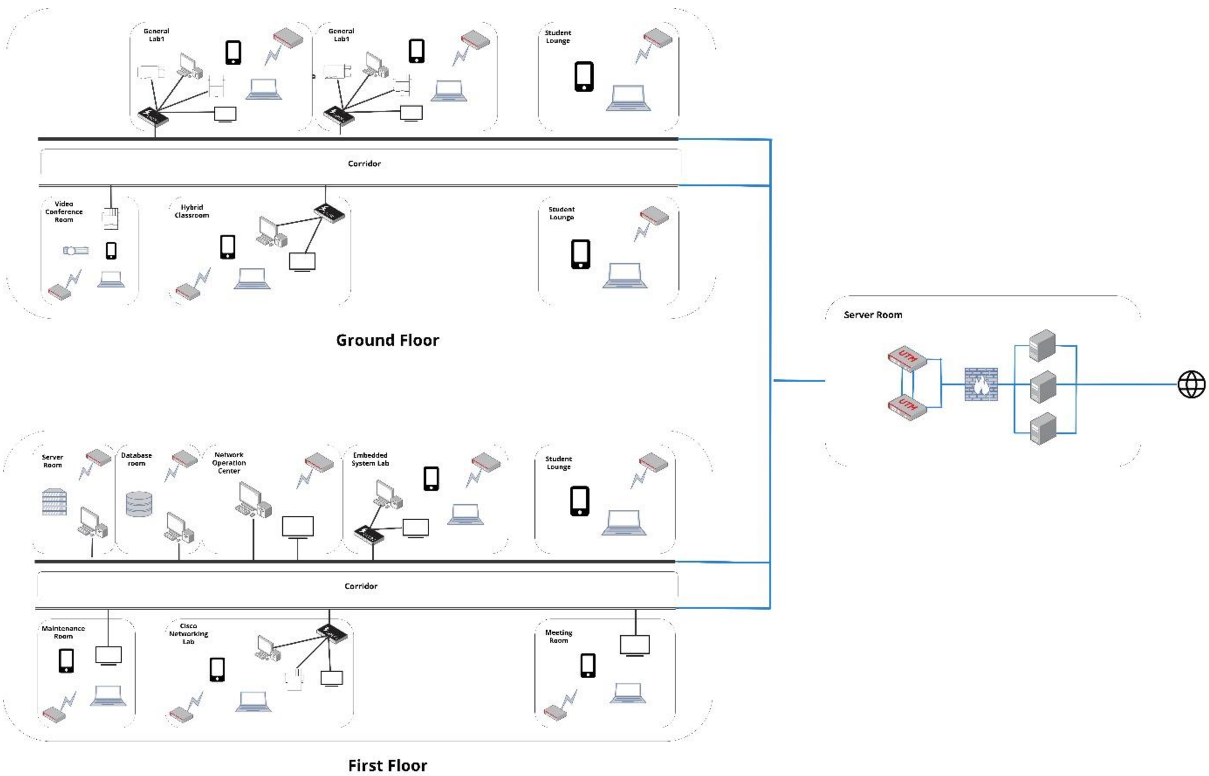

Figure 4.2.1.1: Network Distribution Diagram

The Figure 4.2.1.1 show the network distribution diagram for the ground floor and first floor in our building planning. It illustrates the arrangement of all networking devices and components and how they are connected with each other in the building.

The figure show that each of the room and area will be set up with a wireless access point (WAP). This design is focus on ensuring the fast and efficient communication between each deice while reducing the load on the central router. Moreover, there are five critical and high traffic area which include the two general labs, hybrid classroom, embedded system lab and cisco networking lab will be installed by a switch.

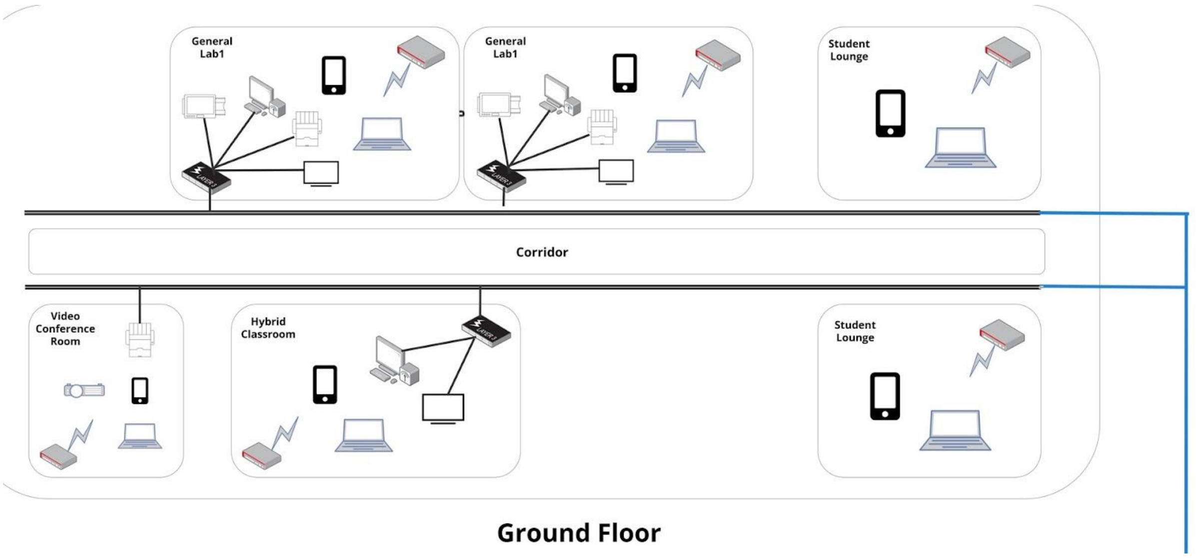

Figure 4.2.1.2: Network Distribution Diagram – Ground Floor

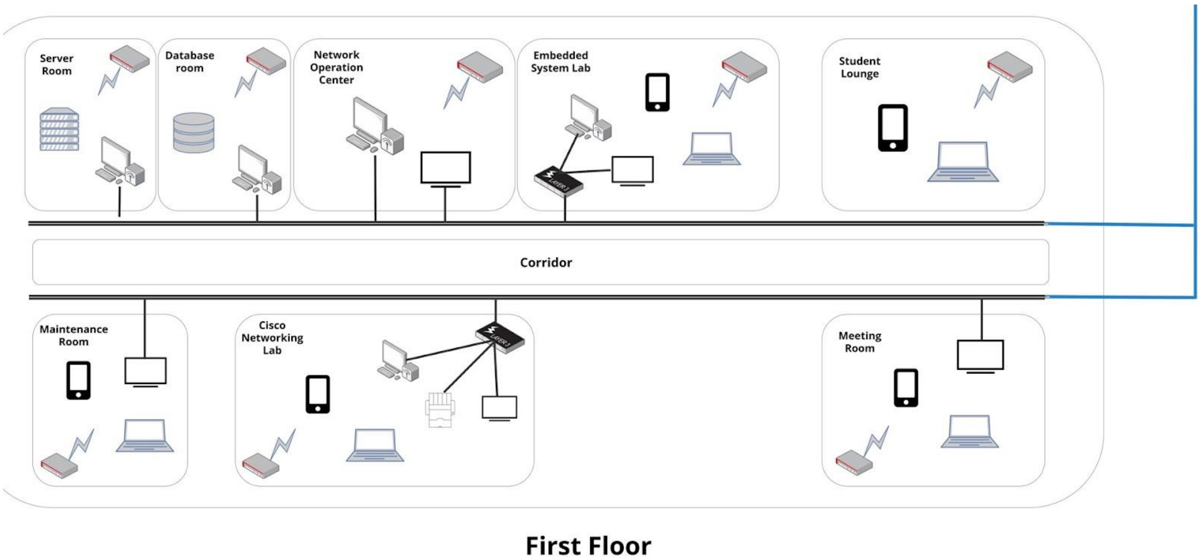

Figure 4.2.1.3: Network Distribution Diagram – First Floor

Figure 4.2.1.2 and Figure 4.2.1.3 show the better view of the network distribution diagram in ground floor and first floor. For each area, including both of the labs, hybrid classroom, video conference room, meeting room, maintenance room and student lounge, will be installed with a wireless access point (WAP) for internet access. There five high traffic areas, include the general labs and hybrid classroom in ground floor, and embedded system labs and cisco networking lab in first floor, will be equipped with a 42-ports switch to handle the high traffic and manage the local communication. The five switches will create subnets with each other to improve network performance and reduce the central router's burden. For the extra ports for each 42-port switch in the five key rooms, they will be connected to other devices. This design can minimize network latency and enhance the transmission speed of data. Therefore, it can ensure the network operates smoothly in the building.

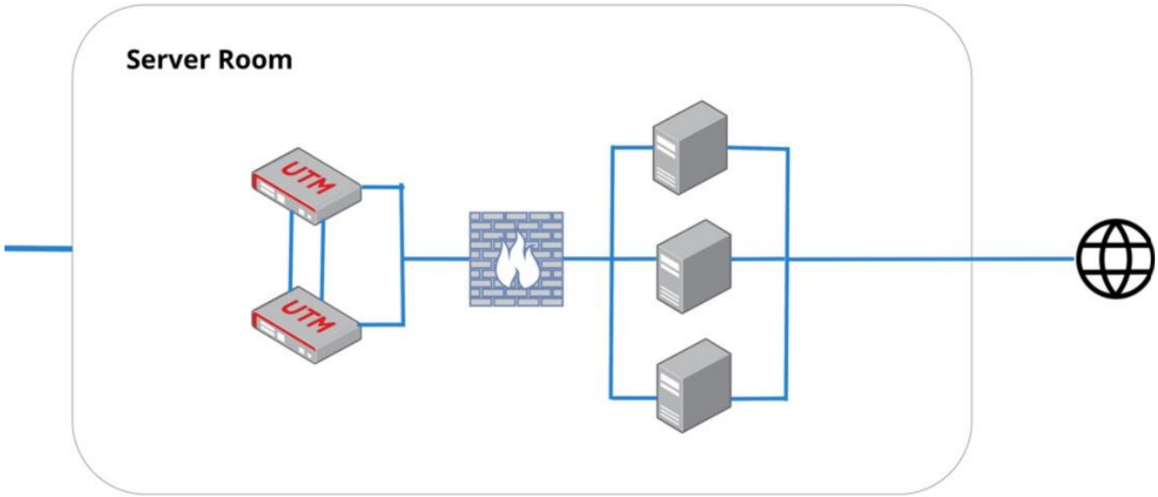

Figure 4.2.1.4: Network Distribution Diagram – Server Room

Figure 4.2.1.4 show the network distribution in server room. The server room consists of two sub-connecting routers, a firewall and three servers. These are the main components for the overall building network functionality and security. The two routers sub-connecting to each other to provide redundancy, which can ensure the network remains operational if one of them fails. The firewall is for protecting the network from potential threads, attacks and also controlling the data traffic. The three servers are function to handle various services and data storage needs across the network. This design ensures high performance, security and seamless internet connectivity in the building.

Figure 4.2.2.1: General Lab

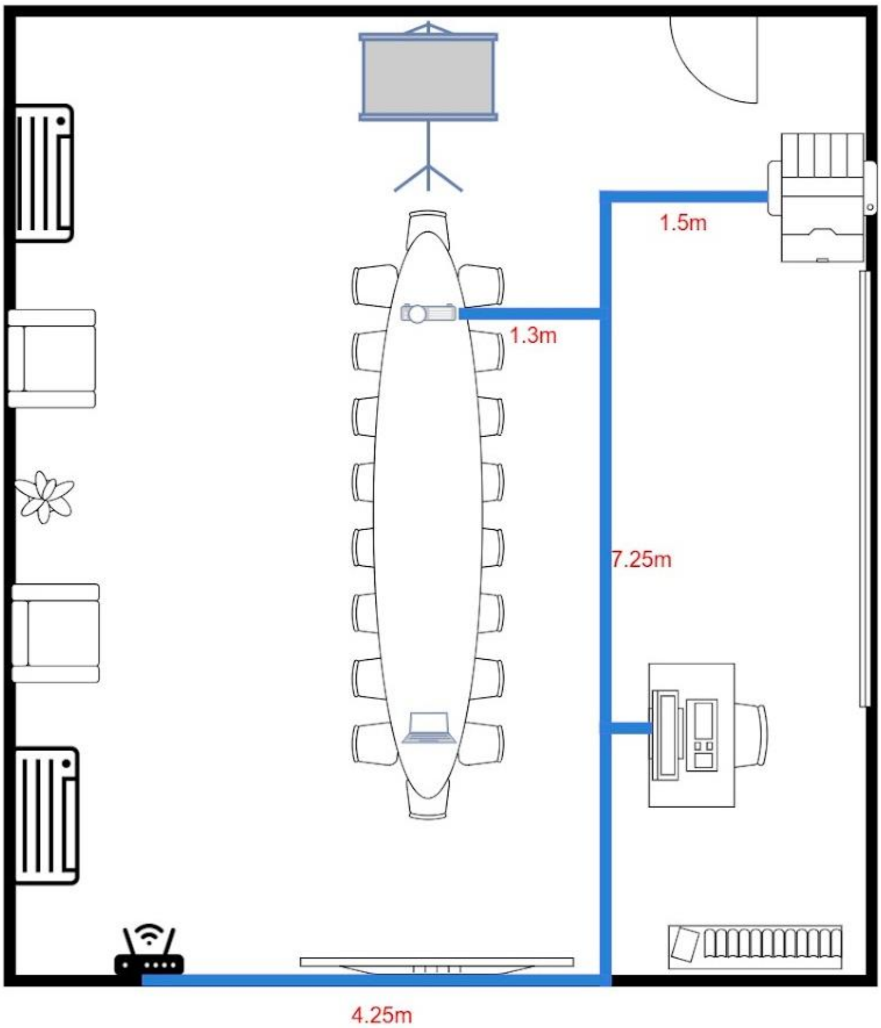

Figure 4.2.2.2: Video Conferencing Room

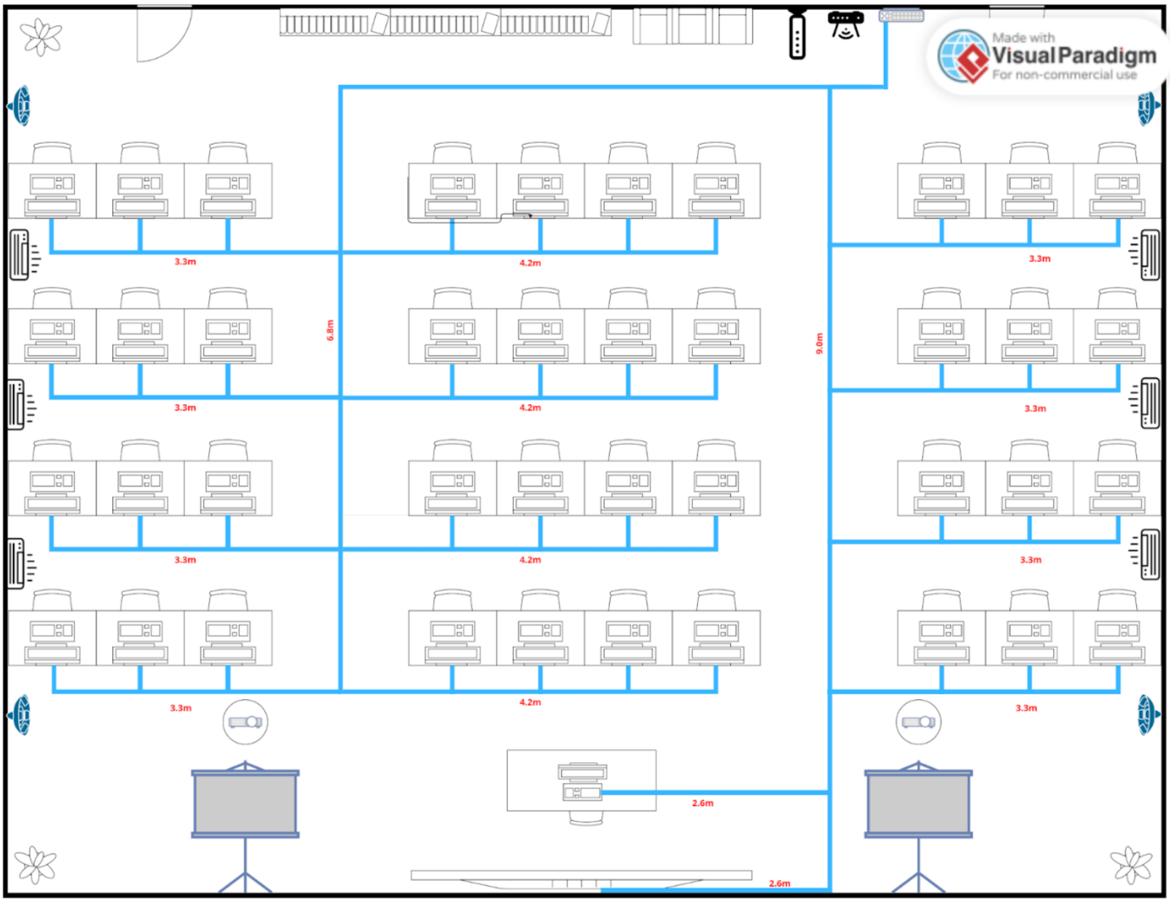

Figure 4.2.2.3: Hybrid Classroom

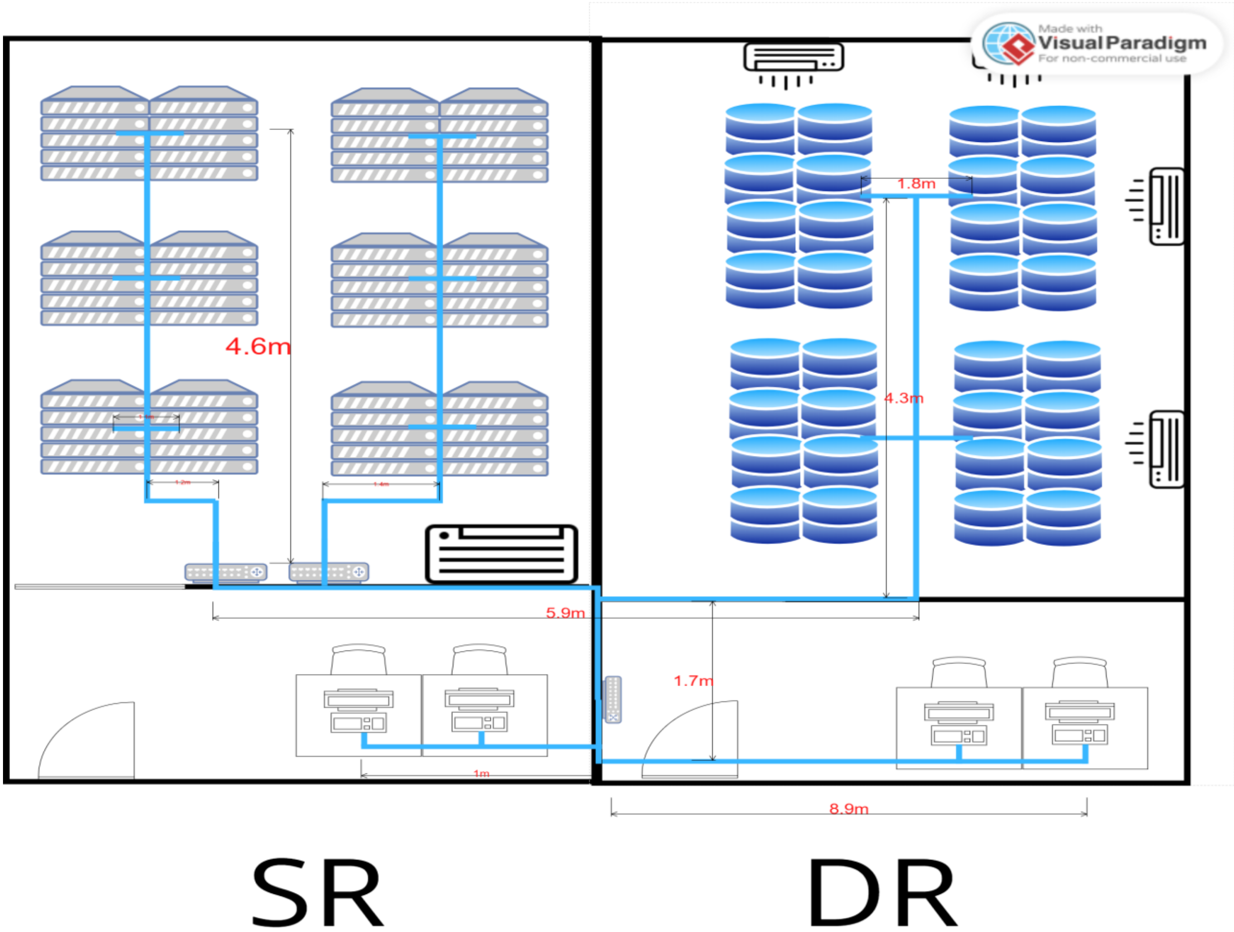

Figure 4.2.2.4: Server room **Figure 4.2.2.5: Database room**

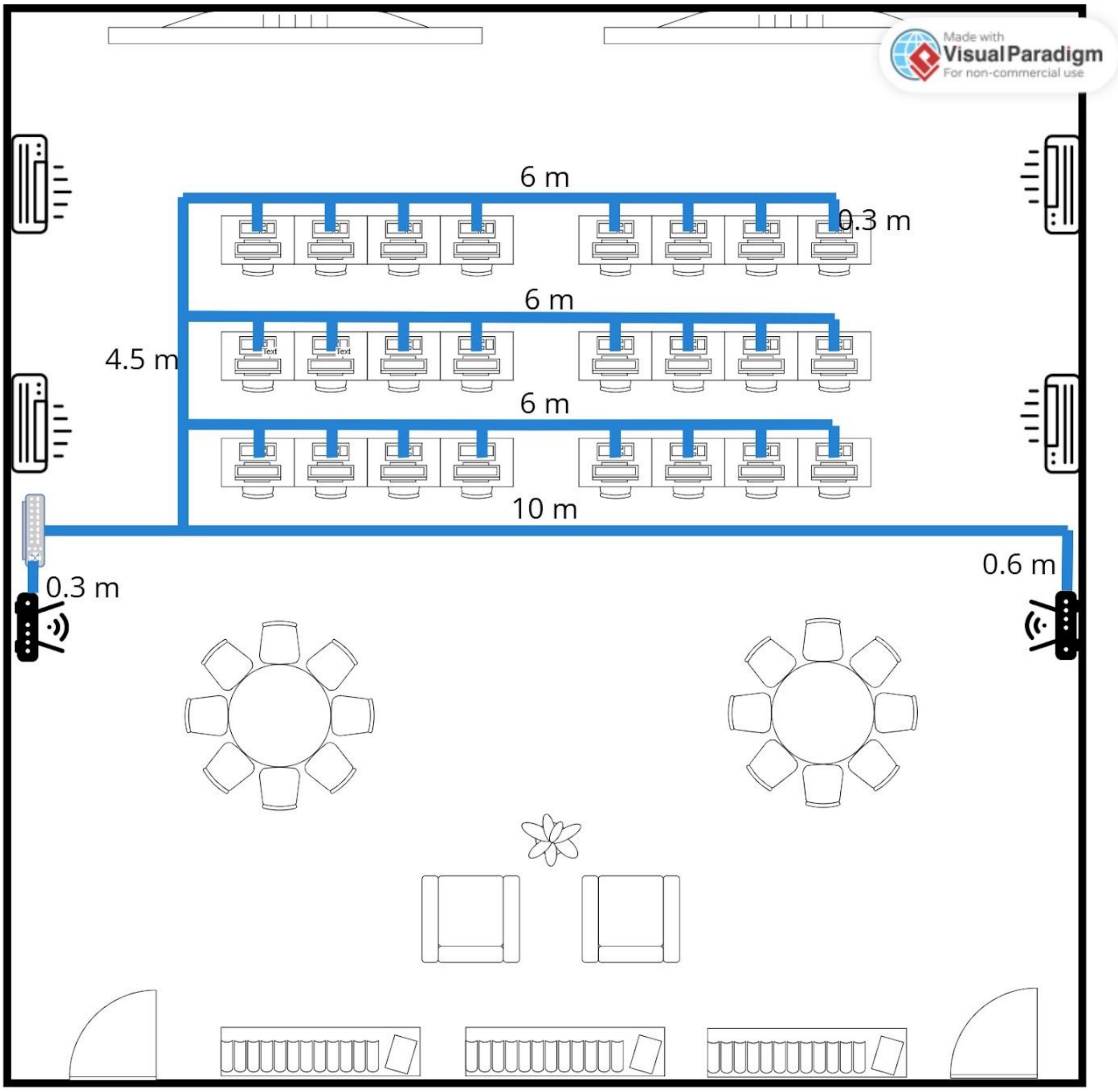

Figure 4.2.2.6: Network Operation Centre

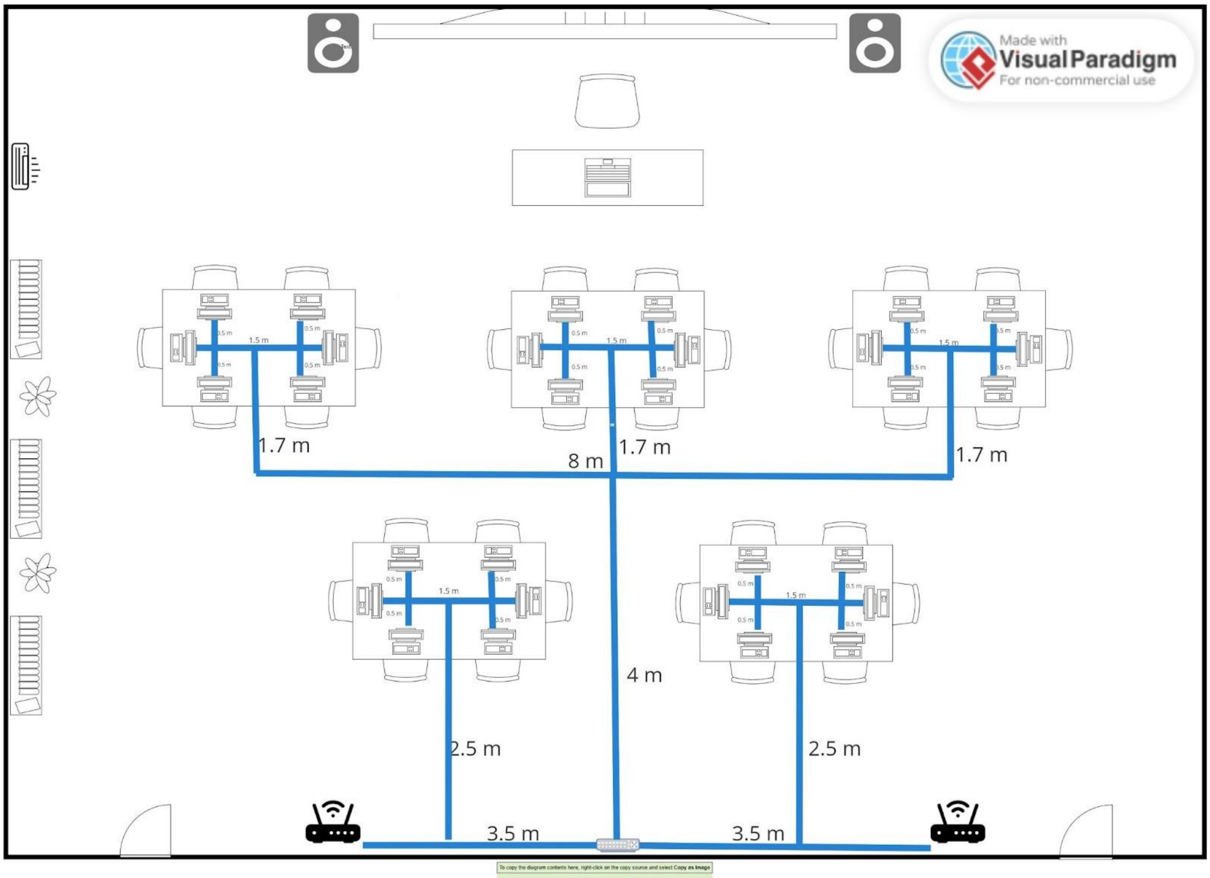

Figure 4.2.2.7: Embedded System Lab

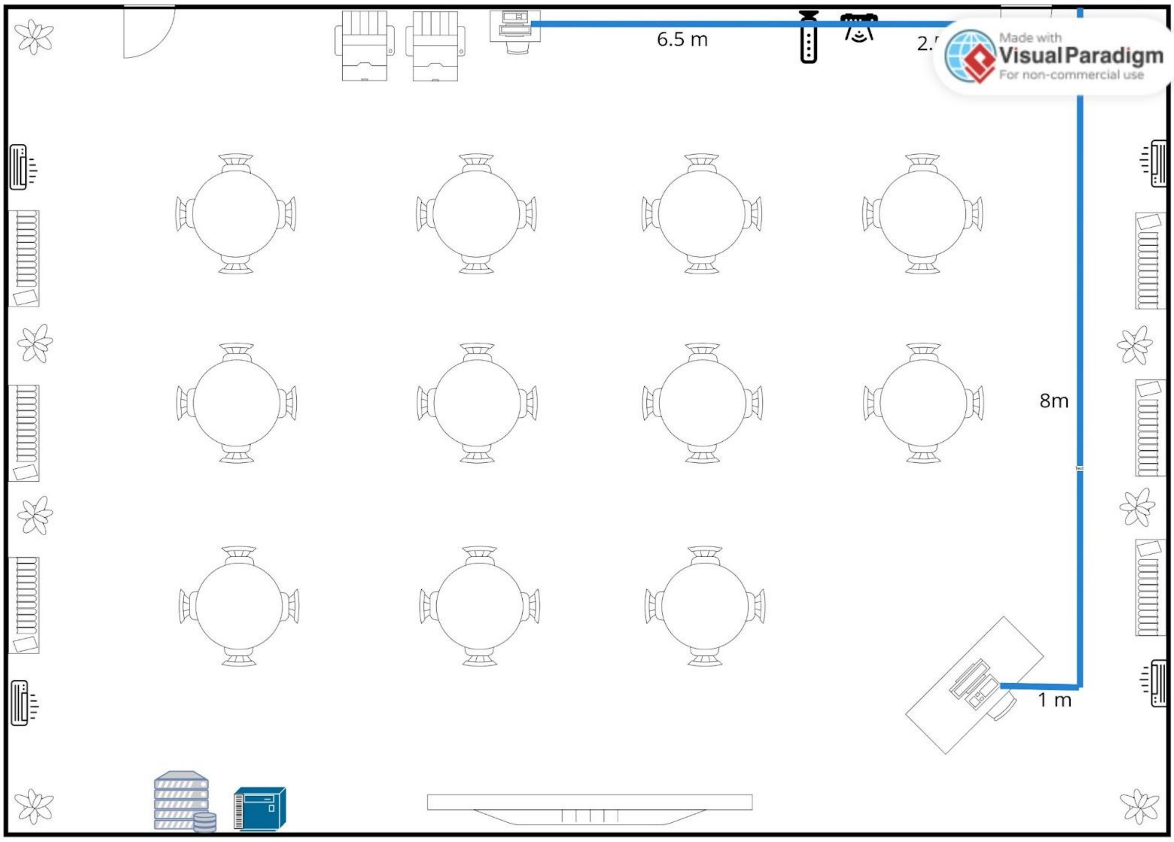

Figure 4.2.2.8: Cisco Networking Lab

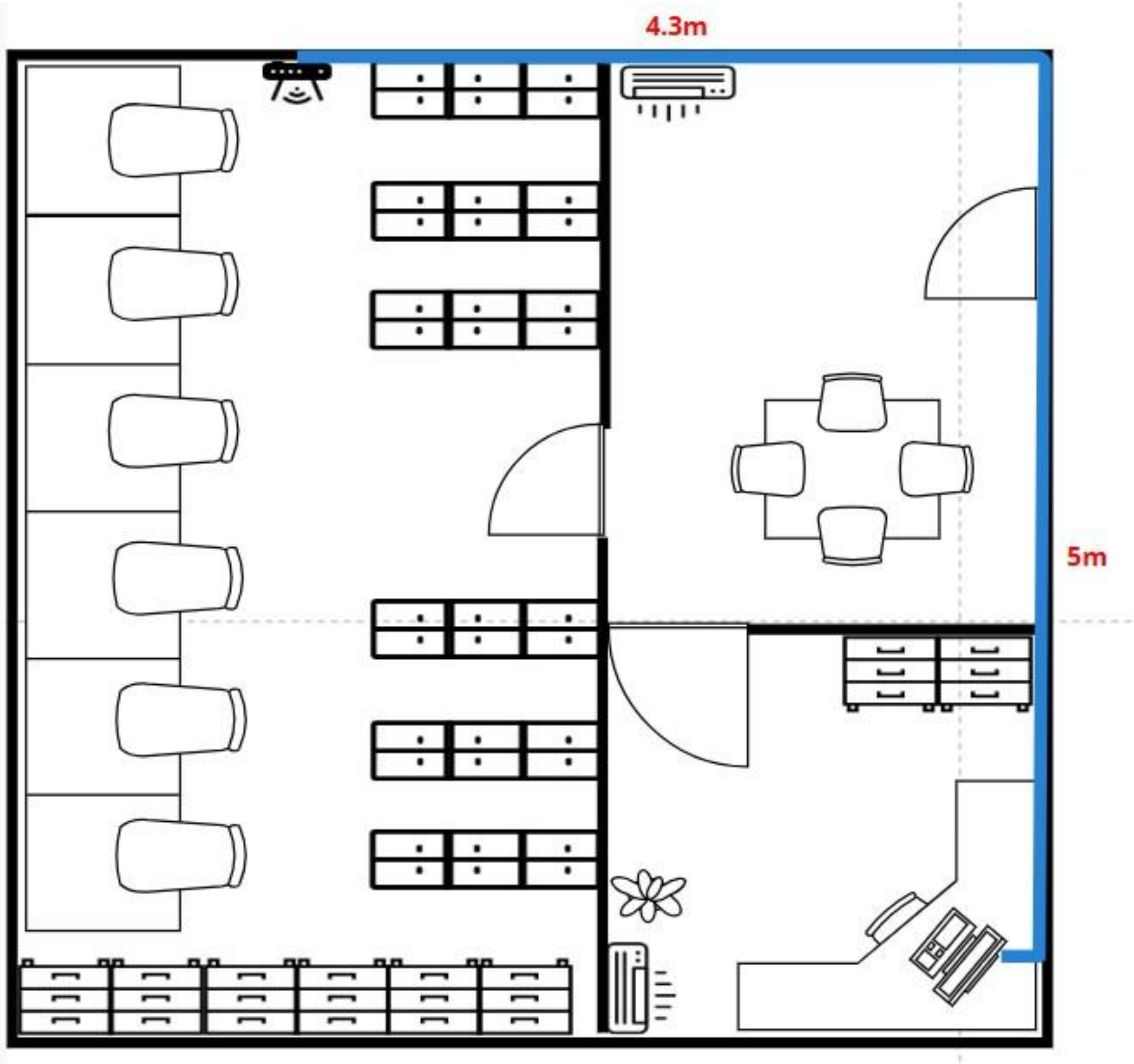

Figure 4.2.2.9: Maintenance Room

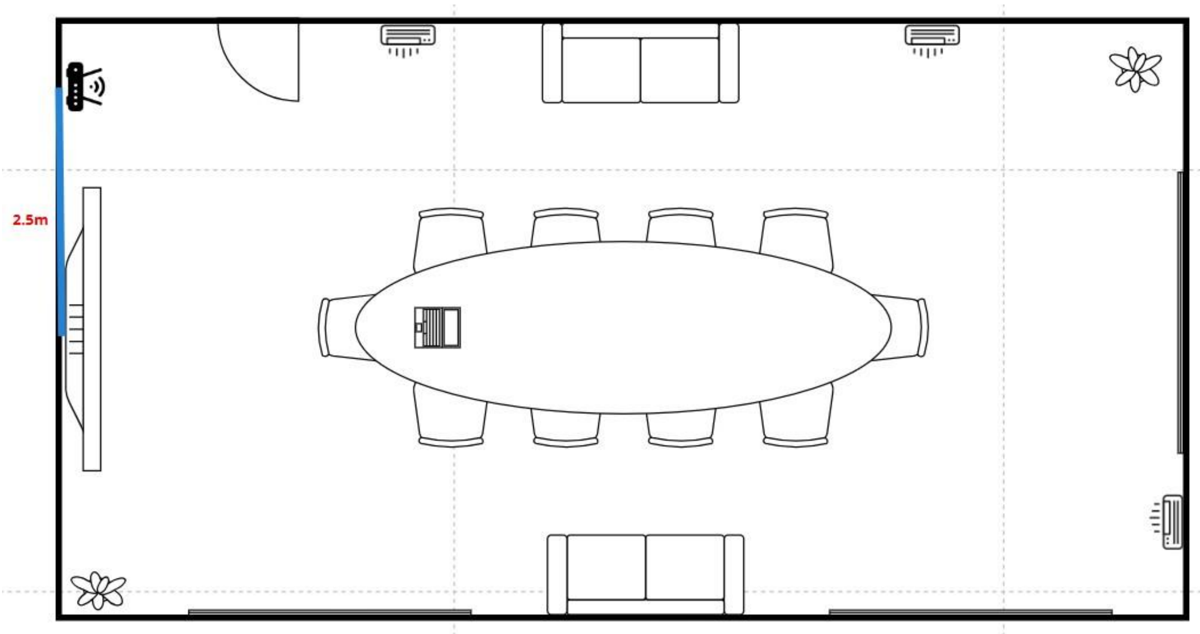

Figure 4.2.2.10: Meeting Room

4.2.3 Floor Diagram

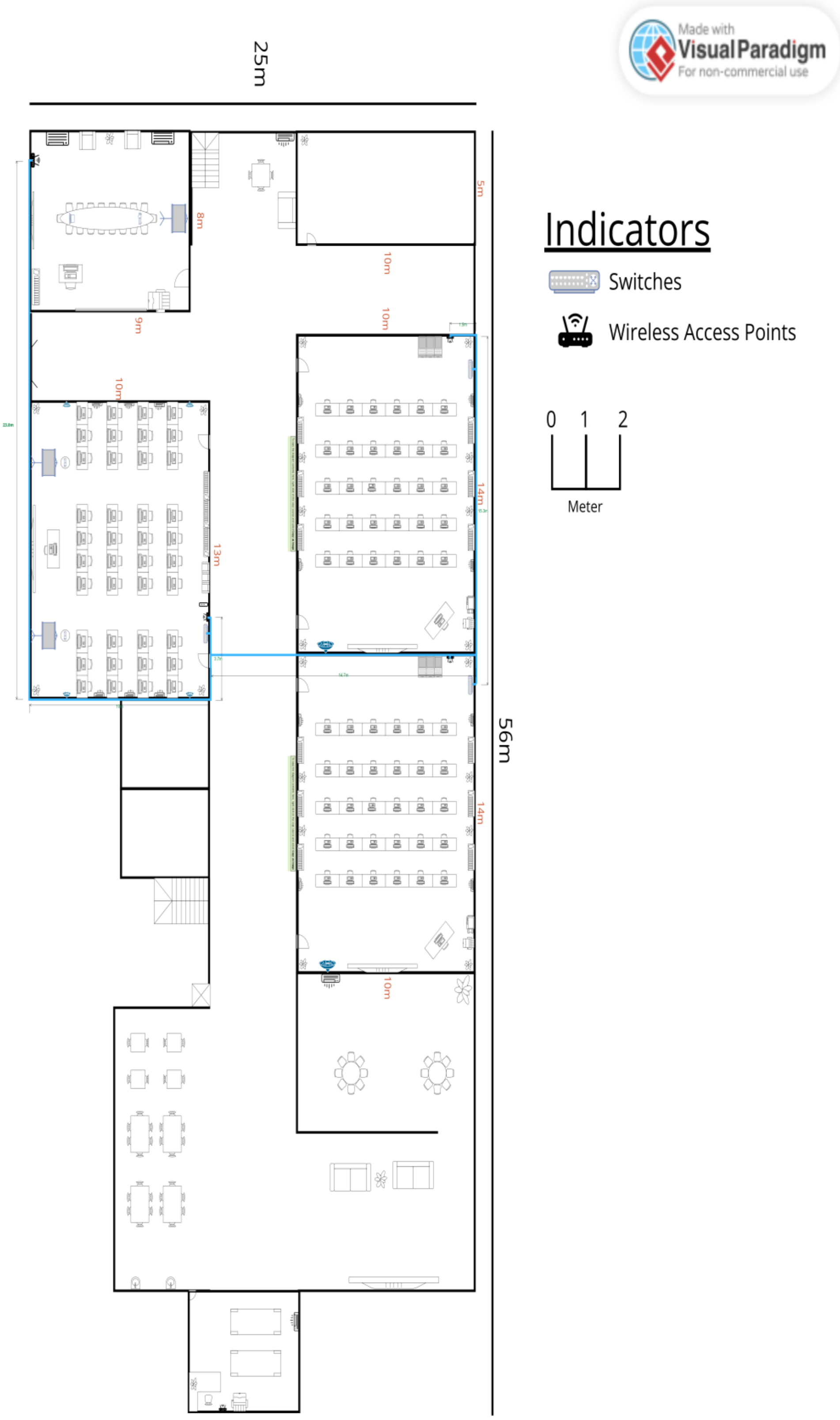

Figure 4.2.3.1 Cable Distribution of First Floor

Figure 4.2.3.2 Cable Distribution of Second Floor

Figures 4.2.3.1 and Figure 4.2.3.2 has provided detailed floor plans showcasing the proposed network layout connecting individual rooms. The network is centralized in the server room, where the router is located, and each room is connected to it through a backbone connection designed to ensure reliable and efficient data transmission. Moreover, the diagrams also highlight the integration of wireless access points across the ground and first floors of the faculty building, demonstrating the comprehensive coverage of the network.

4.3 Number of connections, patch cords and switch ports needed

Area	Number of Switch Ports Needed
General Lab 1	31
General Lab 2	31
Hybrid Classroom	42
Video Conferencing Room	3
Server Room	8
Database Room	8
Cisco Networking Lab	2
Network Operations Center	24
Embedded Systems Lab	30
Meeting Room	1
Maintenance Room	1
Total	181

Table 4.3.1: Number of Switch Ports Needed

Area	Number of Patch Cords Needed
General Lab 1	31
General Lab 2	31
Hybrid Classroom	42
Video Conferencing Room	3
Server Room	8
Database Room	8
Cisco Networking Lab	2
Network Operations Center	24
Embedded Systems Lab	30
Meeting Room	1
Maintenance Room	1
Total	181

Table 4.3.2: Number of Patch Cords Needed

Area	Number of RJ45 Connectors Needed
General Lab 1	62
General Lab 2	62
Hybrid Classroom	84
Video Conferencing Room	6
Server Room	16
Database Room	16
Cisco Networking Lab	4
Network Operations Center	48
Embedded Systems Lab	60
Meeting Room	2
Maintenance Room	2
Total	362

Table 4.3.3: Number of RJ45 Connector Needed

4.4 Identify cable types and length

Area	Length (m)		Total Length (m)	Total Price of Cable Used (m)
	Horizontal	Vertical		
Ground Floor				
Room				
General Lab 1	278.7	15	293.7	3906.21
General Lab 2	278.7	15	293.7	3906.21
Hybrid Classroom	527.4	20	547	7275.10
Video Conference Room	33.55	1.5	35.05	466.20
Ground Floor Room Total			1169.45	15553.72
First Floor				
Room				
Server Room	22.8	4	26.8	356.44
Database Room	58.5	5	63.5	844.55
Cisco Network Lab	20.5	3	23.5	312.55
NOC	187.3	12	199.3	2650.69

ESL	332.6	15	347.6	4623.08
MTR	9.3	2	11.5	152.95
MR	2.5	0	2.5	33.25
First Floor Room Total			674.7	8973.51
Ground Floor				
Path		Length (m)		Price of cables used (RM)
Server room to General Lab 1 and 2		38.6		513.38
Server room to hybrid classrom		33.5		445.55
Server room to Video conference room		68.1		905.73
Ground Floor Room Total		140.2		1864.66
Total - Ground Floor				
Total length of wire to connect the room (m)		140.2		1864.66
Total length of wire in different type (m)		1169.45		15553.72
Total length of wire needed (m)		1467.65		19519.75

First Floor		
Path	Length (m)	Price of cables
Server room to embeded system lab	45.8	609.14
Server room to network operation center	31	412.3
Server room to meeting room	87.4	1162.42
Server room to cicso network lab	54.5	724.85
First Floor Room Total	218.7	2908.71
Total First Floor		
Total length of wire to connect the room (m)	218.7	2908.71
Total length of wire in different type (m)	674.7	8973.51
Total length of wire needed (m)	893.4	11882.22
Total (Ground floor + First Floor)	2361.05	31.401.97

before this we already explain in other section about the distribution plan and device connections. To maximize the connectivity, consideration of cable types is crucial part. So we as a group have discuss, we want to use fiber optic cables and CAT6 twisted cables throughout the building.

For both horizontal and vertical computer cabling in areas like the lab and lecture hall, CAT6a Cables have been chosen due to the ability to support high-speed data transmission. With a bandwidth capacity of up to 550 MHz and data speeds reaching 10 gigabits, CAT6 guarantees smooth performance for high-demand systems, such as coding, cloud computing, and large file transfers. Additionally, these cables support Power over Ethernet (PoE), which makes installation easier by combining power and data delivery into a single cable, cutting costs and clutter. Their strong build guarantees long-term reliability by using thicker copper conductor, Thus improving durabilities and heat dissipating capacity. Futhermore, CAT6 cables are excellent in reducing the crosstalk and electromagnetic interference that caused by natural or human-made sources, offering a reliable and secure connection throughout the building, even in area where signal disruption may occur.

Next, Fiber optic cables have been chosen to connect routers in the server room and link the network to the Internet Service Provider (ISP). These cables use light to transmit data at extremely fast speeds while avoiding signal loss. Fiber optics are perfect for managing massive amounts of data, as they can support bandwidths of up to 10Gbps. Furthermore, their ability to transfer signals across long distances guarantees rapid and trustworthy communication, making them an effective choice for fundamental network connections.

In conclusion, our building requires a combination of CAT6 and fiber optic cables to cover the estimated cable length. This setup ensures smooth connections between devices within the building and a strong, reliable link to the ISP.

Meeting Minute

MEETING MINUTE #1

DATE/TIME	14 Dec 2024 10 pm	
LOCATION	KTDI DSR	
AGENDA	1. Understand the details about task 4.	
	2. Discuss about the network distribution in our floor plan.	
	3. Task distribution for all members	
Meeting MC	NG YU HIIN	
ATTENDANCE		
NAME	TIME	REASON FOR ABSENCE
TAN ZHI MING A23CS0189	2200	
NG YU HIN A23CS0148	2150	
LEE YIN SHEN A23CS0236	2200	
MUHAMMAD NAIM BIN ABDULLAH A23CS0134	2200	

MINUTES			
NO.	ITEM DISCUSSED	IDEA/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Discussion for task 4	• Zhi Ming shared the rubric and instruction about task 4 • All members understand and investigate the task	TAN ZHI MING
2	Suggestion for network distribution	• Yin Shen suggest the network distribution in the floor plan	LEE YIN SHEN
3	Task distribution	- Zhi Ming distributed the task to all member - Yu Hin was assigned to plan for the network distribution of overall floor - Yin Shen was assigned to plan for the network distribution diagram - Naim was assigned to identify the cable types and the total length - Zhi Ming was assigned to do the work areas and network distribution for GL	ALL MEMBER
4	Meeting ended	23:30	ALL MEMBER

MEETING MINUTE #2

DATE/TIME	16 Dec 2024 10 pm	
LOCATION	Online (Google Meet)	
AGENDA	1. Review the current progress for each teammate	
	2. Finalize network distribution diagram	
	3. Check the calculation of cable length	
Meeting MC	TAN ZHI MING	
ATTENDANCE		
NAME	TIME	REASON FOR ABSENCE
TAN ZHI MING A23CS0189	2200	
NG YU HIN A23CS0148	2200	
LEE YIN SHEN A23CS0236	2200	
MUHAMMAD NAIM BIN ABDULLAH A23CS0134	2200	

MINUTES

NO.	ITEM DISCUSSED	IDEA/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Review the progress for each teammate	• All member share their part	All members
2	Finalize network distribution diagram	• All member shared their part using visual paradigm	All members
3	Check the calculation of the cable length	• All members recheck the length calculation to make sure the accuracy	ALL MEMBER
4	Meeting ended	23:00	ALL MEMBER